Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		nitrogen forms NCl_3; phosphorus forms (PCl_3 and) PCl_5 (1) phosphorus is able to expand its octet but nitrogen is not / phosphorus has available *d*-orbitals whilst there are none in the outer shell of nitrogen (1)	2			2		
		(ii)		aluminium has three outer shell electrons so it is able to form three covalent bonds giving six outer shell electrons or an incomplete outer shell or leaving it electron deficient (1) dot and cross diagram of suitable example e.g. $AlCl_3$ and dot and cross diagram of suitable example that has formed coordinate bond e.g. $AlCl_4^-$ / Al_2Cl_6 (1) DO NOT ACCEPT IONIC COMPOUNDS; COORDINATE BOND MUST BE CLEAR	2			2		

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
	(b)			**Indicative content** PHYSICAL STATE 1. carbon dioxide and carbon monoxide have weak forces between separate molecules (so it is a gas) 2. lead has a relatively low ionisation energy so can form positive ions; its oxides are ionic with strong attraction between ions (so it they are solids) ACID/BASE 3. carbon dioxide is a non-metal / covalent oxide so it is an acidic oxide 4. lead(II) oxide reacts with both acids and bases so it is an amphoteric oxide REDOX 5. stability of +2 oxidation state increases down the group / +4 most stable for top of group and +2 most stable at bottom of group / inert pair effect increases down the group 6. carbon monoxide contains carbon in oxidation state +2 whilst +4 is stable, so carbon will be easily oxidised / so it acts as a reducing agent 7. lead(IV) oxide contains lead in oxidation state +4 whilst +2 is stable, so lead will be easily reduced / so it acts as an oxidising agent Relevant equations: • CO as a reducing agent e.g. $CO + CuO \rightarrow Cu + CO_2$ • PbO_2 as an oxidising agent e.g. $PbO_2 + 4HCl \rightarrow PbCl_2 + Cl_2 + 2H_2O$ • CO_2 reacting with a base: $CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$ or $CO_2 + H_2O \rightarrow H^+ + HCO_3^-$ • PbO reacting with a base: $PbO + 2NaOH \rightarrow Na_2PbO_2 + H_2O$ PbO reacting with an acid: $PbO + 2HNO_3 \rightarrow Pb(NO_3)_2 + H_2O$	3	3		6		2

Marks available: AO1, AO2, AO3, Total, Maths, Prac

				5-6 marks The candidate gives at least four relevant points with at least one from two of the three sections, including point 5 and at least two equations. *The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.* **3-4 marks** The candidate gives at least three relevant points with at least one from two sections, including two equations. *The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.* **1-2 marks** The candidate includes at least two relevant points, including one equation. *The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*						
				Question 8 total	**7**	**3**	**0**	**10**	**0**	**2**